

Technical Skills and Tools

- **Programming Languages:** C#, JavaScript, TypeScript, HTML, CSS, Python, Java
- **Frameworks and Libraries:** .NET, WinForms, WPF, ZeroMQ, React, TailwindCSS, Node.js, Express.js, Next.js
- **Databases and ORMs:** MySQL, SQLite, PostgreSQL, Prisma, Sequelize, JPA, Dapper, Entity Framework
- **Cloud Services:** AWS (S3, CloudFront, Elastic Beanstalk, RDS), Supabase, Render, PlanetScale, Vercel
- **Collaboration Tools:** Slack, Microsoft Teams, Zoom, Monday.com
- **Testing and Development Tools:** Git/GitHub, Vitest/Jest, MSTest, Postman, Clerk, Docker, Redis

Professional Experience

Software Engineer	Alluxa	Apr. 2025–Present
Santa Rosa, California		

- Developed a WPF desktop application to integrate a new Bruker FTIR spectrometer into production through TCP/IP which automated data acquisition and allow for very wide spectrum scans for optical filters
- Implemented new stepped and continuous scan methods for a GPIB interfaced Agilent laser spectrometer which reduced scan times by 300% and 150% respectively.
- Automated polarization scan processes involving multi-axis motor control (linear and rotational movement) to position optical components, eliminating manual labor and enabling parallel technician operations.
- Introduced Strategy and Factory design patterns to decouple handling of diverse motor/light-source pairings from core business logic, enabling dynamic selection of scan and movement behaviors and improving overall maintainability.
- Enhanced the Manufacturing Execution System (MES) to meet evolving production requirements and improve workflows across different departments.

Software Engineer I	RheoSense	Aug. 2024–Apr. 2025
San Ramon, California		

- Streamlined performance and actively contributed to resolving customer-related issues in WinForms desktop applications for automated viscometers, enhancing uptime and reliability.
- Integrating pressure calibration into the company's internal web app to automate workflows, enable data upload, and support remote analysis, leveraging RS-232 and USB communication with Fluke pressure controllers and in-house hardware.
- Created a log analyzer tool that reduced troubleshooting time by 80%, increasing service team efficiency and reducing customer downtime.
- Partnered with application scientists to implement product features, directly improving user satisfaction by 15%.
- Supported customers via remote video calls, successfully addressing 95% of reported issues within the first interaction.
- Collaborated with the CEO and upper management to implement Agile-like methodologies and introduce version control best practices, improving development efficiency and team coordination.

Test Engineer II (Software Engineer)	RheoSense	Feb. 2023–Aug. 2024
San Ramon, California		

- Developed and deployed an internal web app integrating 50% of manufacturing processes, saving 20% of labor hours and reducing human error by 60%.
- Developed an internal web app with real-time production metrics, React-based dashboards, and error notifications, streamlining 50% of manufacturing processes and improving team satisfaction by 75%.
- Created decision-assisting algorithms for calibration processes, minimizing errors by 40%.
- Built protected backend RESTful APIs using Express.js, incorporating token-based authentication and input validation to facilitate secure data exchange for web and desktop applications.

- Deployed solutions on AWS infrastructure, leveraging S3, CloudFront, Elastic Beanstalk, and RDS to optimize performance and scalability for remote teams.
- Enhanced cross-functional communication, ensuring seamless project execution across on-site and remote teams.

Engineer I (Production and Test)

RheoSense

May 2021–Feb. 2023

San Ramon, California

- Automated the generation of product certificates with Python scripts utilizing libraries like Pandas and NumPy, reducing manual work by 100 hours annually.
- Conducted comprehensive software and hardware tests for newly developed products, achieving 98% accuracy in pre-launch evaluations.

Education

- **Bachelor's of Science in Chemical Engineering** - University of California, Irvine.

Certificates

- **Amazon Junior Software Developer Professional Certificate** - Coursera **Dec. 2024**